Interior and Exterior Angles of a Triangle

Question Bank - Additional 6.T2 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the angle.

1. The third angle of a triangle with angles of 29° and 88°

63°

180 − 117.

2. The third angle of a right triangle with an angle of 44°

46°

180 − 90 − 44.

3. The exterior angle whose remote interior angles are 33° and 79°

112°

33 + 79.

4. The other remote interior angle, when an exterior angle is 125° and one remote angle is 68°

57°

125 − 68.

For questions 5 through 7, find x and the angles.

5. A triangle's angles are x°, $(2x+15)^\circ$ and $(x+25)^\circ$.

x = 35; 35°, 85° and 60°

4x + 40 = 180.

6. An isosceles triangle's base angles are each $(4x-2)^\circ$ and the third angle is 64°.

x = 15; 58° and 58°

8x − 4 + 64 = 180.

7. An exterior angle is $(5x+2)^\circ$; its remote interior angles are $(3x+10)^\circ$ and 36°.

x = 22; the exterior angle is 112°

2x = 44.

Choose one.

8. A triangle's angles are 25°, 65° and 90°. What is the exterior angle beside the 65°?

- ● **A. 115°**
- ○ B. 65°
- ○ C. 295°
- ○ D. 25°

A

25 + 90.

9. Which equation finds x when an exterior angle is $(3x+6)^\circ$ and its remote interior angles are 54° and $(x)^\circ$?

- ○ A. $3x+6=x$
- ○ B. $3x+6+x=180$
- ○ C. $3x+6+54+x=180$
- ● **D.** $3x+6=54+x$

D

Outside equals the two far ones: x = 24.